Command Broker Commissioning & Joint Training

Standing Up the Command Broker Role at the Vendor–AOO Intersection

Companion Module to the Command Broker Framework and the DBA Series

Module Designation: CB-CT

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Audience: ICS/OT Vendors and Suppliers, and Asset Owner/Operators — Procurement, Engineering, Operations, and Compliance

Parent Document: Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector)

Companion Standard: CB-IB Qualification and Onboarding Standard (CB-IB_Qualification_Standard)

Related: Command Broker Implementation Guide · DBA-VC Vendor Companion · CB-UC-1 BA Integrated Interface

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1. Purpose and Scope

This module governs how a Command Broker is stood up in practice: the commissioning-time and lifecycle procedure by which a qualified human broker is placed in control of a specific product containing an in-scope component — generative AI is the principal case, and FD-LD §4.3 states the scope — at a specific facility, and re-qualified when that product or its model materially changes. It is the connective tissue between obligations the corpus already assigns separately to vendors and to asset owner/operators (AOOs).

The Command Broker is the one role in the corpus that no single stakeholder fully owns. The human broker is the AOO's; the enforcement mechanism, the confirmation interface, the exercise mode, and the audit records are the vendor's; and band classification is a joint determination. The Command Broker Framework establishes the requirement, CB-IB defines broker competency, the Implementation Guide sets AOO governance, and DBA-VC sets vendor product obligations — but none sequences the handoff at which those halves meet. This module supplies that sequence.

Method. This is a standalone companion module. It owns the vendor–AOO seam end to end. Where a requirement is already normative in another corpus document, this module cites it rather than restating it; only the sequencing, the responsibility assignments, the gate criteria, the joint training procedure, and the re-qualification trigger are original and normative here.

- **In scope.** The roles, gates, joint training, and evidence by which a Command Broker is commissioned on a specific product at a specific facility, and re-qualified when the product or model materially changes.
- **Out of scope.** The definition of broker competencies (CB-IB), the Bright Line and criticality-band doctrine (CB-Framework), the confirmation-interface specification (CB-UC-1), and standing vendor product obligations (DBA-VC). This module sequences and connects those; it does not redefine them.

2. Position in the Corpus and Normative Convention

This module sits beneath the Command Broker Framework (parent) and alongside the CB-IB Qualification and Onboarding Standard (companion). It draws operational detail from the Implementation Guide (AOO governance), DBA-VC (vendor product obligations), and CB-UC-1 (confirmation interface).

Conformance convention. A requirement stated as “shall” is binding and introduced by this module. A requirement carried by cross-reference retains the force given it in its home document; this module only places it in sequence. Every requirement names the party accountable for it under the responsibility model in §3. Consistent with DBA-VC §3.1, a vendor’s product characterization does not substitute for the gate evidence required below.

3. Roles and Responsibility at the Seam

The responsibility model below is the spine of the module. R = responsible (does the work); A = accountable (single owner of the outcome and its sign-off); C = consulted; I = informed.

Activity	Vendor	AOO	Mode
Document band classification & config to change band	R	A	Joint sign-off
Deliver enforcement mechanism & confirmation interface	R	I	Vendor
Provide exercise mode (known-erroneous injection)	R	C	Vendor
Staff, qualify & maintain the human broker	C	R/A	AOO
Deliver product-specific interface/exercise walkthrough	R	A	Joint event
Verify audit reconstruction without vendor cooperation	C	R/A	AOO-verified
Model-update notification	R	I	Vendor
Re-qualify broker after material model change	C	R/A	Joint trigger

Governing principle. Per the Command Broker Framework's mechanism-vs-accountability distinction (CB-Framework §2), the vendor may build and operate the enforcement mechanism, but accountability for the discharge of the human decision authority — and therefore for the fitness of whoever discharges it — remains with the AOO. Where that authority is discharged in the moment, it remains with the AOO's named human broker, and the AOO is accountable (A) for every broker-competency outcome in this table. Where the authority is discharged in advance under the pre-authorization mode of FD-BL-D1, the Command Broker discharges it when the pre-authorized envelope and its R2 completeness-and-soundness proof are authored and accepted, and a verified gatekeeper admits each affected action at runtime. The human discharges; the gatekeeper admits. Accountability then rests, under FD-BL-D1 §8, with the AOO's named Command Broker who authored and accepted the envelope and its proof, discharging in advance, together with the asset owner-operator. That person is the Command Broker in the second of the role's two modes, not a second role (FD-BR §4.2). A vendor or supplier that builds or verifies the gatekeeper may not hold that accountability. The responsibility model below is extended accordingly, adding the following activity to the table: "Author, review & accept a pre-authorized envelope and its R2 proof — Vendor C, AOO R/A (named Command Broker, advance mode), Mode: AOO-owned; gatekeeper verification may be vendor-delivered but does not transfer accountability." For each pre-authorized envelope in use at a facility, §3 must name the accountable Command Broker who

authored and accepted it, exactly as it names the accountable broker for each above-line action discharged in the moment.

4. Lifecycle Gates

A Command Broker is commissioned through an ordered sequence of gates. Each gate has entry criteria, defined activities, and an exit condition tied to a specific evidence artifact and a single accountable sign-off. No product executes an above-line action at a facility until the discharge of the human decision authority has been commissioned for the mode in use: for actions discharged in the moment, until the Operational Qualification gate (§4.4) has closed for the individual brokers assigned to them; and for any class of above-line actions discharged in advance under FD-BL-D1, until the Envelope Authorization and Qualification gate (§4.3a) has closed for the envelope governing them.

§4.3a Envelope Authorization and Qualification (pre-authorization mode). Entry. An intended class of above-line actions is proposed for advance discharge under FD-BL-D1; a candidate pre-authorized envelope and its R2 completeness-and-soundness proof have been produced (by the AOO, or built/verified by the vendor for AOO acceptance). Activities. The named Command Broker, holding the advance-mode endorsement of CB-IB §4.1a, reviews and accepts (or rejects) the envelope and its R2 proof over state, trajectory, and rate; verifies that the gatekeeper is the sole path to the process and that the proposer cannot modify or bypass it (R1); and confirms that any change to the envelope, gatekeeper config/limits/logic, or a move toward a protective setpoint is excluded from runtime and routed to requalification (R3). The vendor demonstrates the gatekeeper's determinism and that its verified coverage matches the accepted envelope. Exit & evidence. A signed envelope-acceptance record naming the accountable Command Broker (per §3), the accepted R2 proof, and the verified gatekeeper artifact identifier. This gate closes per envelope. No above-line action is discharged in advance until this gate has closed for the envelope that governs it. Accountable. AOO, through the named Command Broker discharging in advance (FD-BL-D1 §7–§8).

4.1 Procurement

- **Entry.** The criticality band of the intended application has been identified by the AOO (CB-Framework §3.3).
- **Activities.** The vendor provides the band-design statement and the configuration changes required to deploy at a different band (DBA-VC §2.2); the conformance, audit-record, and audit-reconstruction clauses are accepted (DBA-VC §3.1, §3.7).
- **Exit & evidence.** Signed vendor band-design statement and accepted conformance clause on file.
- **Accountable.** AOO, on joint band determination with the vendor.

4.2 Factory Acceptance (FAT)

- **Entry.** Product build complete.
- **Activities.** The vendor demonstrates, action by action, the product’s Bright-Line placement (CB-Framework §3.2) and the operation of the exercise mode that injects known-erroneous recommendations to support broker qualification (DBA-VC).
- **Exit & evidence.** FAT record capturing the demonstrated Bright-Line placements and exercise-mode results.
- **Accountable.** Joint; AOO witnesses and countersigns.

4.3 Site Acceptance / Commissioning (SAT)

- **Entry.** Product installed in the target operating environment.

- **Activities.** On-site verification of the confirmation gesture sequence, staleness/degraded-data flags, and override procedure (CB-UC-1 §§4–5); verification that the AOO can independently reconstruct broker decisions from the audit record without vendor cooperation (DBA-VC §3.7).
- **Exit & evidence.** SAT record plus a passing audit-reconstruction test result.
- **Accountable.** AOO.

4.4 Operational Qualification

- **Entry.** Named brokers assigned; SAT closed.
- **Activities.** The Joint Training Event (§5) is delivered, and each assigned broker demonstrates competency against the CB-IB standard on this specific product.
- **Exit & evidence.** A per-broker competency record (CB-IB) and the training-event evidence package. This gate closes individually, per broker.
- **Accountable.** AOO. No broker operates above the Bright Line until this gate has closed for that individual.

4.5 Model-Update Re-Qualification

- **Entry.** A vendor material-change notice (DBA-VC model-update notification procedure).
- **Activities.** Per §6: classify the change, and where material, re-run the affected Joint Training Event modules and re-qualify affected brokers.
- **Exit & evidence.** Re-qualification record plus confirmation of audit continuity across the change.
- **Accountable.** AOO.

4.6 Periodic Drill

- **Entry.** The review interval or a triggering condition is reached (not less than annually for Highest-band applications, per the Implementation Guide).
- **Activities.** A degraded-mode / adversarial-stress exercise covering the stress categories in CB-Framework §3.6.
- **Exit & evidence.** Drill findings logged; any competency lapse triggers remediation before continued above-Bright-Line operation.
- **Accountable.** AOO.

5. The Joint Training Event

The Joint Training Event is the currently-unspecified centerpiece of the seam: the vendor-led walkthrough of a specific product delivered to the AOO's assigned brokers, paired with the AOO's own process-authority context. It is delivered at the Operational Qualification gate (§4.4) and re-run, for the affected modules, on any material model change (§6). CB-UC-1 establishes that interface training is a component of CB qualification maintenance; this section specifies the event that discharges it.

5.1 Vendor-led content (product-specific)

- The product's Bright-Line placement, action by action (CB-Framework §3.2).
- The confirmation gesture sequence, staleness/degraded-data flags, and override procedure (CB-UC-1 §§4–5).
- Operation of the exercise mode, including a live run in which each broker responds to injected known-erroneous recommendations (DBA-VC).

- How to read and reconstruct the audit record for a brokered decision (DBA-VC §3.7).

5.2 AOO-led content (facility-specific)

- The process-authority context: what the broker is deciding, and the operating limits that bound it.
- Escalation paths and decision-authority boundaries at this facility.
- The broker's non-delegable accountability for the brokered decision (CB-Framework §2).

5.3 Delivery and evidence

The event records attendance, the results of each broker's live exercise-mode run, and a per-broker competency sign-off that feeds the CB-IB competency record. That record is the evidence closing the Operational Qualification gate for each individual.

6. Model-Update Re-Qualification Loop

DBA-VC requires the vendor to notify the AOO, in advance, of model updates, fine-tuning changes, and retrieval-source changes. This module closes the loop that notification opens: it defines when such a change obliges re-training and re-qualification, and it requires audit continuity across the change.

Material-change test. A change is material if it could alter the product's outputs for any class of actions at or above the Bright Line, if it changes the pre-authorized envelope or the gatekeeper's configuration, limits, or logic for any class of actions discharged in advance, or if it changes the confirmation interface, the exercise mode, or the structure of the audit record. A change confined to cosmetic or interface-styling behavior with no effect on above-Bright-Line outputs, on any pre-authorized envelope or its gatekeeper, or on the evidence artifacts is non-material.

Obligation — routed by mode of discharge. The remedy for a material change depends on how the affected above-line actions are discharged. (i) For actions discharged in the moment by a named broker, the affected modules of the Joint Training Event (§5) shall be re-run and the affected brokers re-qualified before the updated product's above-line actions are again discharged in the moment. (ii) For any class of above-line actions discharged in advance under FD-BL-D1, a change to the pre-authorized envelope itself — or to the gatekeeper's configuration, limits, or logic, or a move of the pre-authorized envelope toward a protective setpoint — is categorically excluded from runtime and shall route to offline requalification producing a new verified artifact (FD-BL-D1 §4, R3): a new envelope and a fresh R2 completeness-and-soundness proof, re-accepted by the named Command Broker, discharging in advance, through the Envelope Authorization and Qualification gate (§4.3a) before the changed envelope governs any action. Broker re-qualification is not the remedy where the Command Broker discharged in advance; requalification of the verified artifact is. There is no live in-field learning of the gatekeeper, and no window of unverified operation between artifacts. Audit continuity shall be preserved so that pre-change and post-change decisions — whether discharged in the moment or in advance — remain reconstructable from a coherent record, including which envelope version and verified artifact governed each pre-authorized action.

Open decision. Whether the material-change threshold above is maintained here or migrated into DBA-VC as a vendor-side definition is a corpus-placement decision for the author; the substance is intended to be identical wherever it lives.

7. Evidence and Artifact Mapping

Every gate resolves to an evidence artifact already required elsewhere in the corpus. This module imposes no new evidentiary burden; it sequences existing artifacts and assigns their sign-off.

Gate / requirement	Evidence artifact	Home document
Procurement	Band-design statement; accepted conformance clause	DBA-VC §2.2, §3.1
Factory Acceptance	Bright-Line placement & exercise-mode results	CB-Framework §3.2; DBA-VC
Site Acceptance	Audit-reconstruction test result	DBA-VC §3.7; CB-UC-1 §§4–5
Operational Qualification	Per-broker competency record	CB-IB; this module §5
Model-Update Re-Qualification	Re-qualification record; audit-continuity attestation	DBA-VC; this module §6
Periodic Drill	Degraded-mode exercise findings	CB-Framework §3.6; Implementation Guide

8. Conformance and Misrepresentation

Each requirement in this module binds the party named accountable for it in §3. Consistent with CB-Framework §3.2 and DBA-VC §3.1, a vendor's characterization of a product — for example as offering "turnkey autonomy" or "operator-free optimization" — does not substitute for the gate evidence required here, and does not by itself close any gate. No product containing an in-scope component executes an above-line action at a facility until the gate for the mode in use has closed — the Operational Qualification gate (§4.4) for the brokers who discharge in the moment, or the Envelope Authorization and Qualification gate (§4.3a) for the envelope governing actions discharged in advance — and re-closed after any material change under §6.

9. Foundational References

Command Broker family instruments are cited in this document by designation, without version tokens. The current version of each is stated in CB-REG, which is the authoritative register for this family.

DBA-EN series instruments are cited in this document by designation, without version tokens. The current version of each is stated in DBA-EN-SF §1.4, which is the authoritative register for this series; a version-bearing citation would duplicate that register and drift from it between revisions. Resolve any series instrument named in this document against DBA-EN-SF §1.4. External regulatory references and supporting documents retain their version and date tokens.

Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector). Roxey, T.E. Eclectic Technologies, 2026. [Parent: broker requirement, Bright Line, criticality-band model, mechanism-vs-accountability distinction.]

CB-IB Qualification and Onboarding Standard (CB-IB_Qualification_Standard). Roxey, T.E. Eclectic Technologies, 2026. [Companion: normative broker competency, onboarding, and accountability.]

Command Broker Implementation Guide. Roxey, T.E. Eclectic Technologies, 2026. [AOO duty-of-care, training, and governance.]

DBA-VC Vendor Companion. Roxey, T.E. Eclectic Technologies, 2026. [Vendor obligations: joint band classification, exercise mode, model-update notification, evidentiary package, audit reconstruction.]

CB-UC-1 BA Integrated Interface. Roxey, T.E. Eclectic Technologies, 2026. [Confirmation interface, staleness flags, interface training as qualification maintenance.]

Roxey, T.E. “Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers.” Version 29. Eclectic Technologies, 2026. [Primary risk-landscape source for the three-tier framework and the Bright Line concept.]

Revision History

Version	Date	Description
0.1	July 2026	Initial draft. Establishes the standalone Commissioning & Joint Training module (Option A): responsibility model at the vendor–AOO seam, lifecycle gates, the Joint Training Event, the model-update re-qualification loop, evidence mapping, and conformance. Sequences and cross-references existing corpus requirements without restating them.
0.2 (cite refresh)	Jul 2026	Citations refreshed to current corpus versions: CB-Framework v0.2, Command Broker Implementation Guide V0.2, CB-CT v0.2, DBA-VC v0.8. Mechanical version-reference pass; no substantive content change. (Held pending decision: the NERC/BES-Variant TAF reference and the ISA Parent Framework V1.2 label.)
0.3	September 2026	Citation-strip pass, version-pin pass Tier 4, and a house-style repair. Ten version tokens stripped: the cover parent, companion and related lines, and five entries of Section 9. Command Broker family citations take the CB-REG lead-in; the DBA-VC citation is a DBA-EN series citation and takes the series lead-in resolving to DBA-EN-SF §1.4, so both lead-ins stand at the head of Section 9, differing in scope only. Separately, this document's header and footer parts were present in the package but empty, so it has carried no running header, no version footer and no page numbers since issue; both are built here to house style. The version increment is the second: the preceding entry records a revision taken as 0.2 (cite refresh), and the gap is left visible.
0.4	October 2026	Conformed to the October 2026 Edition of the Foundational Definitions (10.5281/zenodo.23117612) under Project 0’s work order of 3 October 2026. The “named human envelope author” of §3, §4.3a and §6 is named as what Decision C1 makes it: the named Command Broker discharging in advance, holding the advance-mode endorsement of CB-IB §4.1a (FD-BR §4.2; FD-BL-D1 §7, §8). The responsibility model is otherwise unchanged; the added activity keeps its R/A assignment and its parenthetical is renamed. The Command Broker discharges and the gatekeeper admits. §4, §6 and §8 no longer say a product operates above the Bright Line, which places a product rather than an action, and §8’s closing gate condition now covers the advance mode, which it

Version	Date	Description
		omitted. §1 states the scope by FD-LD §4.3. §6 names the pre-authorized envelope where v0.3 said “operating region” (FD-BL-D1 R3; FD-EV).